

What is DeFi?

Learn about decentralized finance systems.

Fidelity Viewpoints



Key takeaways

Decentralized finance, or DeFi, is an emerging network of peer-to-peer financial services that uses blockchain technology to facilitate lending, borrowing, staking, and trading.

Advocates believe DeFi may revolutionize the traditional financial system by cutting out intermediaries like banks and credit card companies.

Skeptics cite regulatory uncertainty, technical complexities, and potential risks of hacks and fraudulent activity.

DeFi advocates have ambitious goals—many of them wish to rewire traditional financial systems like banking and credit card payments. They believe [blockchain](#) technology can help replace most, if not all, of the predominant parts of the financial establishment.

How do [DeFi](#) advocates plan on achieving this? And why should even the most passionate supporters be cautious right now? Let's dive in.

What is DeFi and how does it work?

DeFi is an umbrella term for apps, platforms, and organizations that enable users to lend, borrow, stake (we'll cover more on what staking is shortly), and trade crypto assets.

These functions are accomplished with [smart contracts](#) and [blockchain technology](#). Smart contracts are digital contracts that can be programmed so that a predetermined action happens once certain requirements are met.

Here's an example of how this works. Let's say Jane wants to take out a loan. Typically, she might apply for one through her bank. With [DeFi](#) [smart contracts](#), however, Jane can connect directly with a lender without the need for a bank. That's because all the logistics of the loan, including the terms and the ability to track repayments, can be programmed into the smart contract.

DeFi aims to apply this way of operating to as many aspects of traditional finance as possible. Advocates think it can provide an open, transparent, and efficient alternative to the established financial system. Anyone with a smartphone or computer can take part, no matter where they're located. There are no third-party entities that decide who can or can't participate.

Most DeFi platforms are known as protocols, a term which describes the functions and rules of the service. Users mostly conduct transactions with DApps (short for decentralized applications), which are software-based apps built upon blockchains, most commonly on the Ethereum network. In simple terms, they're like the apps on your smartphone or computer, but they operate with blockchain technology.

DeFi has gained significant attention in recent years, but the community's aspiration to disrupt traditional finance faces skepticism for a variety of reasons. Critics warn that many DeFi projects are highly speculative and volatile. Furthermore, unlike traditional finance, DeFi platforms do not offer insurance—users potentially risk losing their investments in the event of a hack or a smart-contract failure. Additional drawbacks, according to skeptics, include regulatory concerns and scalability limitations.

Centralized finance vs. decentralized finance

As you learn about DeFi, you may come across the distinction between centralized finance and decentralized finance. Centralized finance—sometimes referred to as “CeFi” or “TradFi” by the crypto community—describes the world of traditional banks, brokers, insurers, and credit card companies.

Many DeFi supporters think CeFi has shortcomings, including creditworthiness requirements, high fees, lengthy wait times for wire transfers, limitations on access to funds, and other factors. They believe the remedy, of course, is DeFi—an ecosystem where developers are also able to create new products and services without the need for approval from centralized entities.

Yet this also means that DeFi lacks many of the built-in protections that existing centralized finance systems have. These include benefits like regulatory oversight, customer support, and more user-friendly interfaces. In addition, blockchains which host DeFi platforms can become congested, which may lead to higher transaction fees and slower confirmation times.

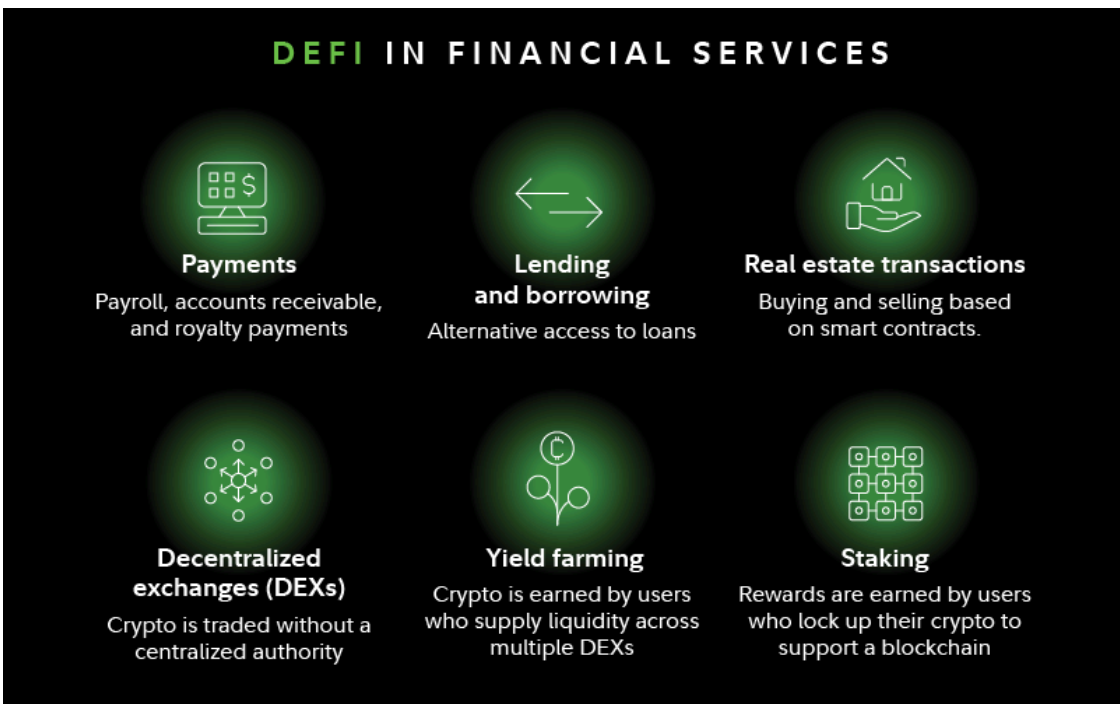
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What are some DeFi use cases?

DeFi is currently being applied to a variety of financial services. A few of these include:



For illustrative purposes.

DeFi supporters envision a future where smart contracts can automate payroll, accounts receivable, and royalty payments to ensure accuracy and timeliness. They believe this will also help cut down on the administrative costs required to manage these items.

For skeptics, the decentralized nature of DeFi comes with a lot of risks. For example, unlike centralized financial services, DeFi protocols don't have customer support. There's no central team that can resolve disputes or reverse transactions in case of error. This may make using DeFi for significant financial activities (i.e., payroll) riskier or less practical than traditional methods.

Decentralized exchanges (DEXs)

How do DEXs operate? Most crypto trading occurs via centralized exchanges (CEXs). Decentralized exchanges (DEXs), however, aim to allow trading without a centralized authority. They run on smart contracts, which supporters believe can make trading more trustworthy and allow participants to stay anonymous.

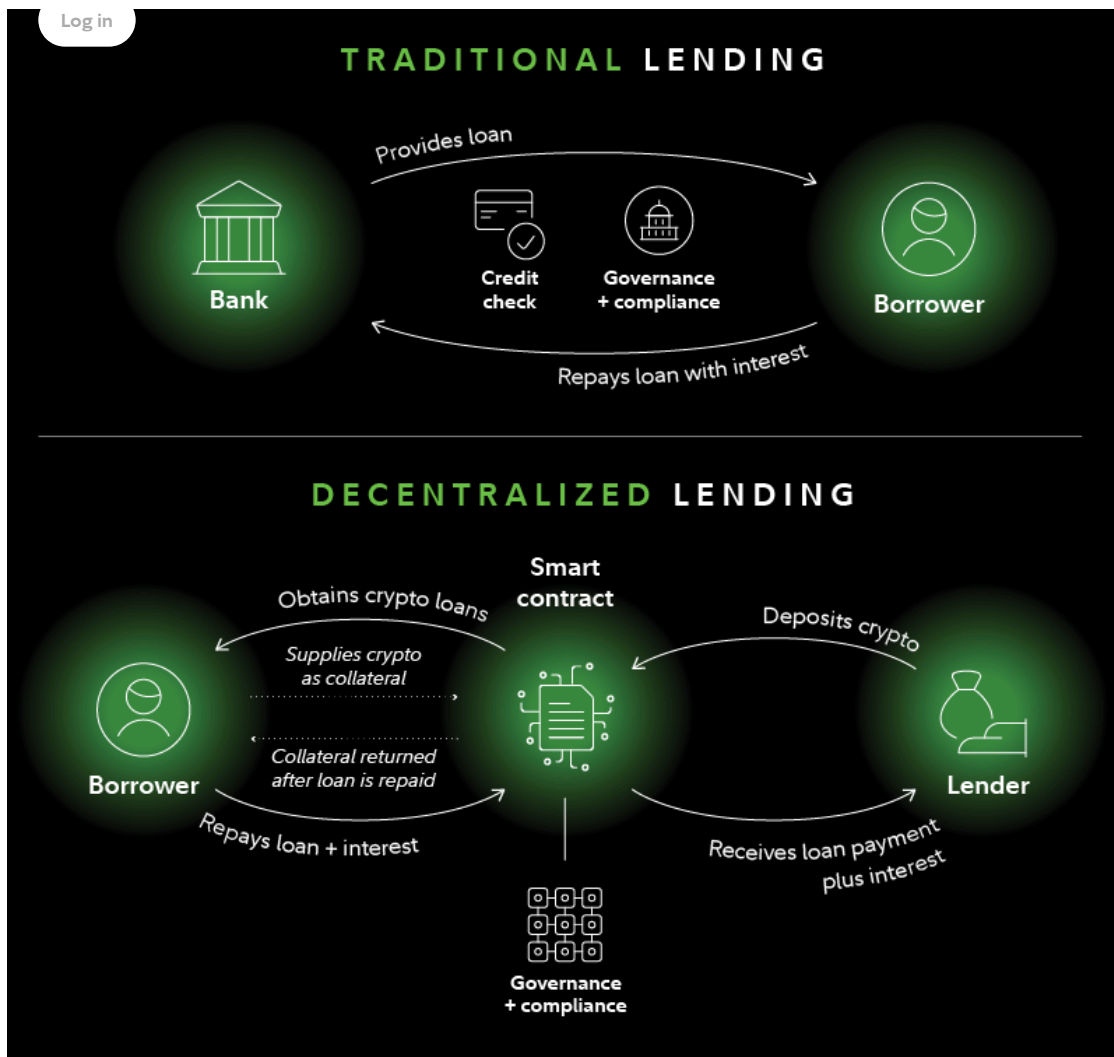
However, while decentralization may offer greater privacy, a main trade-off is that there is regulatory uncertainty, which can lead to greater risk of scams and frauds. For example, smart contracts are a relatively new technology and can potentially face technical vulnerabilities. In the last few years, several high profile DeFi protocols have been hacked for over 9 figures in losses. Skeptics believe it's not worth putting your financial assets on the line assuming these kinds of risks.

Also, since DEXs have fewer participants than CEXs, users may experience lower trading volumes and lower liquidity, in addition to potential price disparities. Furthermore, custody of assets is also directly linked to users' wallets—instead of an account on a CEX—leading to potential security risks.

Lending and borrowing

Advocates believe DeFi can make it easier for more people to access lending, as approval doesn't rely on many of the strict criteria required by traditional lenders. DeFi lending involves supplying crypto to protocols which, in turn, can be borrowed in exchange for interest. Borrowers must provide collateral in the form of other crypto assets, which are sometimes worth more than the value of the amount they want to borrow. Why would a borrower do this? Hypothetically it supplies crypto liquidity to a borrower who may not wish to sell the specific crypto assets that are being put up for collateral.

As previously noted, DeFi lending platforms don't have the typical consumer requirements that exist in traditional finance, which can be a double-edged sword, as those requirements are often implemented to protect the consumer. Additionally, crypto volatility may create unfavorable conditions for both borrowers and lenders.



For illustrative purposes.

Yield farming

Yield farming allows participants to earn crypto by supplying liquidity across multiple DEXs to increase returns. Users can choose to lock up their crypto in accounts known as liquidity pools, which helps make trading on DEXs run more smoothly.

However, skeptics note that DeFi products are currently complicated to use, requiring a deeper, more sophisticated knowledge of the crypto landscape and its unique ins and outs. Without technical knowledge of how smart contracts work, less experienced users may be at greater risk of making mistakes, and the slightest errors may result in losing access to their assets forever. In addition, yield volatility on certain platforms can potentially lead to rapid devaluation of returns.

Staking

Staking cryptocurrencies is a process that involves committing or “locking up” your crypto assets for specific periods of time to support a blockchain network and confirm transactions. Staking allows users to earn crypto rewards on the amount they lock up. This helps blockchains because the crypto that’s locked contributes to the network’s stability and security.

Still, given that some protocols require staked crypto to be locked up for a predetermined time, users may experience reduced liquidity and flexibility with their assets. In addition, due to the potential security vulnerabilities of smart contracts, stakers may be at risk of losing their locked-up funds.

Real estate transactions

Traditionally, buying and selling real estate can involve multiple intermediaries, which can make the process expensive and slow. DeFi supporters hope that smart contracts can open the door for faster and more cost-efficient transactions that don't need third parties. They also believe they can be used to verify ownership of the property more effectively than existing methods.

Given DeFi is still in its infancy, using it for large transactions like real estate may pose certain challenges, including security risks with smart contracts. In addition, the concept of tokenized real estate on blockchain is fairly new and may create obstacles for buyers and sellers. The property market is highly regulated and tax structures may create greater complexity for transactions, especially across different geographic regions.

What to consider before participating in DeFi

Before using any DeFi protocols, first [learn how they work](#). As mentioned earlier, many of these platforms can be complex and intimidating for new users. For example, most DeFi activities require the use of a [crypto wallet](#), which can be a complicated tool to figure out in and of itself. Remember: Even the slightest mistake could result in losing access to your assets for good. So be sure you're confident about how a platform functions before using it.

Also, note that crypto may be more susceptible to market manipulation than securities, and DeFi platforms may be more vulnerable to security concerns than centralized finance platforms. Crypto holders and DeFi users do not benefit from the same regulatory protections applicable to registered securities. Crypto is also not insured by the Federal Deposit Insurance Corporation or the Securities Investor Protection Corporation, meaning you should only [buy crypto](#) or interact with DeFi protocols with an amount you're willing to lose.

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